



American International University-Bangladesh (AIUB)

Faculty of Science & Technology (FST)
Department of Computer Science

MID-TERM PROJECT REPORT
SUMMER 2025-2026

Project Title	Multi-user Expense Tracker
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Course / Section	Programming In Python [Sec: D]
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Date of Submission	17/07/26

1. Executive Summary

This project is a simple “Expense Tracker” program made with Python. The main goal of this project is to help people keep track of their daily income and expenses. More than one person can use this program so we call it a multi user expense tracker. Each user can add their income and expenses, see their balance, and delete entries if they make a mistake. The program saves all the data in a file so the information is not lost when the program is closed. We used basic Python ideas like classes, lists, and tuples to build this project. We also imported a library called NumPy to calculate things like total expenses and average expenses. This project helps people understand where their money is going and how much they are spending every day.

2. Analysis Methodology

We used Python to build this project. We used NumPy to calculate the total expenses, average expenses, highest expense, and lowest expense. We used classes and objects to organize our code. We also used a file handler to save the data so it does not get lost.

3. Program Output and Interface

When the program runs for the first time, it asks the user to enter a username and then displays the main menu showing the user's name, current balance, budget limit, and all available options like Add Entry, Delete Entry, View History, and more.

```
===== WELCOME =====
Available users:
1. New user
Select user: 1
Enter username: Robin

===== BUDGET TRACKER =====
User: Robin
Balance: 0.0 BDT
Budget Limit: 0.0 BDT

* 0.00 BDT remaining.

1. Add Entry
2. Delete Entry
3. View History
4. Set Budget
5. Search by Category
6. Statistics
7. Save
8. Delete user
9. Quit

Select option: █
```

When the user selects option 3, the program displays a complete transaction history showing all income and expense entries with their dates, categories, and amounts, followed by the updated balance and remaining budget.

```
Select option: 3

History:

Income:
1. 17-07-2026 +20000.00 BDT

Expense:
1. 18-07-2026 [Transport] -500.00 BDT
2. 19-07-2026 [Food] -4000.00 BDT

===== BUDGET TRACKER =====
User: Robin
Balance: 15500.0 BDT
Budget Limit: 15000.0 BDT

* 10500.00 BDT remaining.
```

When the user selects option 5, the program shows all used categories and lets the user search by a specific category, then displays all expenses under that category with the total amount spent.

```

Select option: 5

--- Search by Category ---
Categories used: Food, Transport
Enter category to search: Food

Expenses in category 'Food':
  - 19-07-2026: 4000.00 BDT
Total for category 'Food': 4000.00 BDT

```

When the user selects option 6, the program uses NumPy to calculate and display the total expense, average expense, highest entry, and lowest entry from all recorded expense data.

```

Select option: 6

Statistics:
Total Expense: 4500.00 BDT
Average Expense: 2250.00 BDT
Highest Entry: 4000.00 BDT
Lowest Entry: 500.00 BDT
Categories Used: ('Food', 'Transport', 'Bills', 'Health', 'Shopping', 'Education', 'Other')

```

When the program is run again after closing, it automatically loads all previously saved users and their data, proving that the file handler is working correctly and no data is lost even after the program is closed.

```

===== WELCOME =====
Available users:
1. Robin
2. New user
Select user: 1

===== BUDGET TRACKER =====
User: Robin
Balance: 15500.0 BDT
Budget Limit: 15000.0 BDT

* 10500.00 BDT remaining.

1. Add Entry
2. Delete Entry
3. View History
4. Set Budget
5. Search by Category
6. Statistics
7. Save
8. Delete user
9. Quit

Select option: █

```

The program supports multiple users at the same time, as seen here where both Robin and Joe are available as existing users, and a new third user can also be added, proving that each user has their own separate data stored in the system.

```

===== WELCOME =====
Available users:
1. Robin
2. Joe
3. New user
Select user: 3
Enter username: █

```

4. Key Findings

1. Each user can add their own income and expenses separately.
2. The program can calculate the total income and total expense automatically.
3. NumPy helped us find the average, highest, and lowest expense very easily.
4. The program saves all data in a file so nothing is lost when the program is closed.
5. The user can delete a wrong entry if they made a mistake.

5. Limitations

Though our project works well but it has a few small things that can be improved in the future. First, we did not add a GUI which means the user has to type everything in the “Terminal” instead of clicking buttons. Second, our project does not have any password system, so any person who opens the program can see other users’ data.

6. Conclusion

We built a fully working multi user expense tracker using Python with a Command Line Interface (CLI), and even though it is a simple project, it actually solves a real-life problem of tracking daily money in an organized way.

Final Submission Checklist

- ✓ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- ✓ I attached the project report as a separate file.
- ✓ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusion.
- ✓ My code uses NumPy library meaningfully.
- ✓ All bracketed placeholder text and helper notes were replaced or deleted before submission.